Martin Moreno Jara

Computer Science and Engineering Student | Bogota, Colombia
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PROFESSIONAL SUMMARY

Computer Science and Engineering student, focused on backend systems, system design and architecture. Comfortable with Python (FastAPI, Flask) and REST API. Seeking an internship starting in 2027.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, C++ (Arduino), SQL
Frameworks and Libraries: FastAPI, Flask, Express.js, React, Node.js
Tools and Infrastructure: Git, GitHub, Linux, Docker, Excel, Firebase, Google Cloud Platform, Prometheus, Grafana
Databases: MongoDB, SQLite
Concepts: System Design, REST API Design, Microservices, Distributed Systems, Computer Networks, HTTP, MQTT, TLS/SSL, Pub/Sub Messaging
Spoken Languages: Spanish (native), English (advanced), German (intermediate)

EDUCATION

Universidad Nacional de Colombia | Bogota, Colombia
B.Sc. in Computer Science and Engineering (Ingeniería de Sistemas y Computación) | Feb 2022 - Jun 2027 (Expected)

Relevant coursework: Software Engineering and Architecture, Databases, Data Structures and Algorithms, Computer Networks, Operating Systems.

PROJECTS

Intelligent Pot | firmware | dashboard | May 2026 - Jul 2026
C++ (Arduino), ESP32-S3, MQTT, HiveMQ

- Designed an IoT planter built on an ESP32-S3 that samples five environmental sensors (soil moisture, temperature, humidity, ambient light, air quality), displays readings locally, and publishes telemetry over TLS-secured MQTT to a HiveMQ Cloud broker.
- Defined the MQTT topic contract between device and dashboard, using a retained state topic so a client connecting late receives the device's current configuration immediately.
- Closed the control loop over pub/sub: the dashboard publishes to a command topic to trigger irrigation, adjust the watering threshold, and configure lighting and audio on the device remotely.
- Owned system design and integration across the embedded, messaging, and web layers, reconciling firmware and dashboard into a single working product documented for external readers.

Runpath | auth service | api gateway | Sep 2025 - Dec 2025
Python, FastAPI, Express.js, Nginx, Firebase, Docker, Google Cloud Platform

- Built a microservices infrastructure with an Express.js API Gateway and an Nginx reverse proxy, centralizing request routing across services.
- Implemented an authentication backend with FastAPI and Firebase, handling identity management and access control.
- Configured encrypted communication between internal services using self-signed certificates to protect data in transit.
- Containerized services with Docker and deployed them to Google Cloud Platform.

Mascotas Bogota | publications API | telemetry | Apr 2025 - Aug 2025
Python, Flask, MongoDB, Prometheus, Grafana

- Developed the core RESTful API of a community pet-tracking platform in Python and Flask, covering CRUD operations for user publications.
- Built an observability pipeline with Prometheus and Grafana, exposing real-time metrics for application health and backend performance.